

SOC Automation Project

Alert Enrichment & Automated Ticket Drafting Workflow

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Environment	Wazuh 4.7.5 OpenSearch Freshdesk
Classification	Project Writeup

1. Executive Summary

This report documents the design, build, and testing of an end-to-end SOC automation workflow built on a homelab Wazuh SIEM environment. The project was conceived to reduce manual analyst workload by automatically enriching security alerts with contextual intelligence and routing them to a ticketing system, while suppressing known-good behaviour.

The completed workflow covers the full journey from raw Microsoft 365 log ingestion through to a formatted, AI-enriched Freshdesk ticket landing in the analyst review queue. Where alerts match known user behaviour in a maintained Knowledge Base, they are automatically suppressed and logged. Where they do not, the alert is enriched with IP intelligence from AbuseIPDB, contextual analysis from a locally-hosted LLM, and delivered as a structured ticket.

The solution was validated against a synthetic dataset of 100 simulated users across AAD, Exchange, and OneDrive workloads, and is designed to be deployable across a multi-tenant MSP environment serving approximately 50 clients.

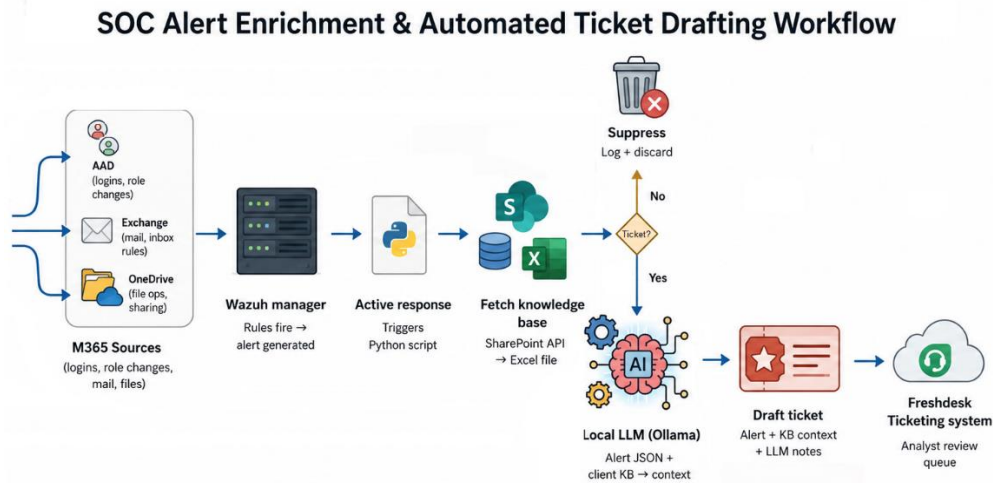


Figure 1 - Project overview flow diagram

2. Project Objectives

- Replicate the production Wazuh/OpenSearch/M365 environment in a homelab for safe rule and automation development.
- Build a synthetic M365 data injection pipeline producing realistic AAD, Exchange, and OneDrive events across 100 simulated users.
- Develop and validate custom Wazuh detection rules targeting logins from non-trusted countries (Russia and China/Hong Kong).
- Implement an active response script that performs Knowledge Base lookups against a SharePoint-hosted Excel file to determine suppress or escalate decisions.
- Enrich escalated alerts with AbuseIPDB IP intelligence and LLM-generated analyst context via a locally-hosted Ollama model.
- Deliver formatted, analyst-ready tickets to Freshdesk automatically on escalation.
- Design the solution to be portable across client tenants with no hardcoded client-specific logic.

3. Solution Architecture

3.1 High-Level Workflow

The automation pipeline consists of seven sequential stages:

Stage 1	M365 log ingestion via Wazuh Office 365 module (AAD, Exchange, OneDrive)
Stage 2	Filebeat ships alerts.json to OpenSearch via ingest pipeline with GeoIP and ASN enrichment
Stage 3	Wazuh analysisd evaluates rules and fires alerts on matching conditions
Stage 4	Active response triggers kb_enrichment.py on rules 100100 and 100101
Stage 5	Script fetches Knowledge Base from SharePoint (OneDrive) and performs user/IP/country lookup
Stage 6	If known behaviour: suppress and log. If unknown: query AbuseIPDB and Ollama LLM
Stage 7	Formatted ticket created in Freshdesk with alert data, IP intel, and LLM analyst notes

3.2 Infrastructure Components

SIEM	Wazuh 4.7.5 (single-node, Ubuntu 22.04)
Index/Search	OpenSearch (wazuh-indexer)
Log shipper	Filebeat 7.10.2 with custom ingest pipeline
GeoIP enrichment	MaxMind GeoLite2 City, Country, ASN databases
Knowledge Base	Excel (.xlsx) hosted in Microsoft OneDrive/SharePoint
LLM	Ollama (llama3.8b) running locally on the Wazuh server
IP Intelligence	AbuseIPDB REST API
Ticketing	Freshdesk (REST API v2)

3.3 Ingest Pipeline

The Filebeat ingest pipeline was modified from the default Wazuh configuration to include GeoIP enrichment from the MaxMind City database and ASN organisation name lookup from the ASN database. A Painless script processor merges the two GeoLocation objects so that both geographic fields (city_name, country_name, region_name, location) and ISP context (organization_name) are available at root level on each indexed document.

Field path: data.office365.ClientIP is the source IP field for all Office 365 GeoIP lookups.
GeoLocation.country_name and GeoLocation.organization_name are the enriched output fields.

4. Detection Rules

4.1 Rule Design

Two custom Wazuh rules were developed in the 100000+ ID range (reserved for local custom rules) to detect logins from non-trusted countries. Rules are evaluated by the Wazuh manager at alert generation time and fire before ingest pipeline enrichment. As such, they operate on the raw socket event fields rather than the enriched OpenSearch document.

Rule ID	100100
Description	Login from non-trusted country (Russia)
Condition	office365.Operation = UserLoggedIn AND office365.ClientIP matches known Russian IP ranges
Level	10 (High)
MITRE ATT&CK	T1078 - Valid Accounts
Groups	office365, geolocation, non_trusted_country

Rule ID	100101
Description	Login from non-trusted country (China / Hong Kong)
Condition	office365.Operation = UserLoggedIn AND office365.ClientIP matches known Hong Kong/China IP ranges
Level	10 (High)
MITRE ATT&CK	T1078 - Valid Accounts
Groups	office365, geolocation, non_trusted_country

4.2 Rule Validation

Both rules were validated using wazuh-logtest with raw event payloads. Rule 100100 was confirmed to fire correctly on Russian IP addresses and produce an alert at level 10. End-to-end testing via the Wazuh socket confirmed alerts land in OpenSearch under the correct rule ID and are visible in Discover and the dashboard.

5. Knowledge Base

5.1 Structure

The Knowledge Base is maintained as a Microsoft Excel file (.xlsx) hosted in SharePoint/OneDrive and accessible via the Microsoft Graph API using a registered Azure AD application. The file contains one row per user with the following columns:

Column A	data.office365.UserId - The user's full email address (primary key)
Column B	GeoLocation.country_name - Comma-separated list of known login countries
Column C	data.office365.ClientIP - Comma-separated list of known IP addresses
Column D	data.office365.DeviceProperties.DisplayName - Known device display names

5.2 Design Decisions

The Knowledge Base uses a single row per user rather than one row per behaviour entry. This simplifies script logic (single dictionary lookup) and makes the LLM prompt more concise. Multiple known countries, IPs, and devices are stored as comma-separated values within each cell.

The Knowledge Base for the test environment covers 100 simulated users. The majority are configured with United Kingdom login countries, reflecting a realistic UK-based client. A small number of users have EU or US countries listed. Only two users (gabi.wood@contoso.com and david.scott@contoso.com) have Russia listed as a known country, simulating users with legitimate travel or remote working requirements.

No users have Hong Kong/China listed as a known country, meaning any China-origin login will always escalate to the analyst queue.

5.3 Caching

The active response script caches the Knowledge Base locally at /var/ossec/etc/google/kb_cache.xlsx with a one-hour TTL. On cache expiry, the script attempts to download a fresh copy from SharePoint via the Graph API. If the download fails, the script falls back to the stale cache to ensure continuity of service. This approach means Knowledge Base updates made by analysts take effect within one hour without requiring script redeployment.

6. Active Response Script

6.1 Overview

The `kb_enrichment.py` active response script is deployed at `/var/ossec/active-response/bin/kb_enrichment.py` and runs as the `wazuh` user. It is triggered by Wazuh's active response mechanism on rules 100100 and 100101. The script receives the full alert JSON via stdin in Wazuh 4.x format (`parameters.alert`) and processes it through the following stages:

6.2 Processing Stages

- Alert parsing - Extracts `UserId`, `ClientIP`, `DeviceProperties`, and rule description from the alert JSON.
- Knowledge Base lookup - Loads the cached KB and performs an exact match on `UserId`. Checks whether the triggering IP and country are listed as known behaviour for that user.
- Suppress/Escalate decision - If IP and country are known: suppress and log. If country is known but IP is not: escalate (new device or location change). If neither is known: escalate.
- AbuseIPDB enrichment (escalate path only) - Queries the AbuseIPDB API for the triggering IP address, returning malicious confidence score, ISP, usage type, associated domain, country, and total abuse reports.
- LLM enrichment (escalate path only) - Sends the alert context, KB entry, and AbuseIPDB results to Ollama (llama3.8b) with a structured prompt requesting a risk rating and recommended investigation steps.
- Ticket creation - Builds a formatted HTML ticket body and POSTs it to the Freshdesk API, setting priority based on the AbuseIPDB malicious score.

6.3 Ticket Format

Tickets delivered to Freshdesk follow a consistent analyst-ready format:

```
Login Date/Time: May 7, 2026 @ 10:30:00
User Account: liam.ferguson@contoso.com
Expected Login Location(s): United Kingdom
Unusual Login Location: Russia
Managed device used: No
Name of device used: DESKTOP-ABC123
IP Address: 77.88.55.60
```

Intelligence:

- Malicious Score: 0%
- ISP: Yandex LLC
- Usage type: Data Center/Web Hosting/Transit
- Domain associated with IP: yandex[.]ru
- Country of Origin: Russia
- Reported for malicious activity: No

Analyst Notes (AI-generated):

```
Risk: Medium - Login from Russia on unmanaged device not in KB.
1. Verify with user whether travel to Russia was expected.
2. Check for concurrent sessions from other locations (impossible travel).
3. Review recent mailbox and OneDrive activity for this user.
```

6.4 Priority Mapping

AbuseIPDB Score >= 50%	Urgent (Priority 4)
AbuseIPDB Score >= 20%	High (Priority 3)
AbuseIPDB Score < 20%	Medium (Priority 2)

7. Test Data & Validation

7.1 Synthetic Data Generation

A Python script (`inject_m365_alerts.py`) was developed to generate and inject realistic synthetic M365 events directly into the Wazuh alerts pipeline via `alerts.json`. The script simulates 100 users across three workloads:

AzureActiveDirectory	25% of events - logins, failed logins, role changes, password resets, account disables
Exchange	40% of events - email sends, mailbox logins, inbox rule creation, hard deletes, SendAs
OneDrive/SharePoint	35% of events - file access, uploads, downloads, sharing, anonymous link creation

IP addresses were drawn from a weighted pool reflecting realistic UK-based client traffic: 60% UK ISPs (BT, Virgin Media, Sky), 25% EU and US, with a small proportion of Russian and Hong Kong IPs to trigger the custom detection rules. Events include realistic device properties, user agent strings, operation types, and timestamps weighted toward business hours.

7.2 Pipeline Validation

Events injected via the `alerts.json` route are processed by Filebeat and the ingest pipeline, landing in OpenSearch with correct GeoLocation enrichment, enabling dashboard visualisations and rules to function identically.

Events injected via the Wazuh socket are evaluated by `analysisd` and matched against detection rules before being written to `alerts.json` and shipped to OpenSearch. This route is used for active response testing.

7.3 Test Results

Rule 100100 (Russia)	Confirmed firing via wazuh-logtest and socket injection. Alert visible in Discover and Security Events.
Rule 100101 (China)	Confirmed firing via wazuh-logtest.
KB suppress - known user	<code>gabi.wood@contoso.com</code> from <code>77.88.55.60</code> correctly SUPPRESSED. Log entry confirmed.
KB escalate - unknown user	<code>liam.ferguson@contoso.com</code> from <code>77.88.55.60</code> correctly ESCALATED. Log entry confirmed.
AbuseIPDB enrichment	Confirmed returning malicious score, ISP, domain, and report count.
Ollama LLM enrichment	Confirmed generating risk rating and investigation steps (llama3.8b).
Freshdesk ticket creation	Confirmed ticket created via API with formatted HTML body and correct priority mapping.

8. Future Enhancements

- Impossible travel detection - Custom rule firing when the same UserId authenticates from two geographically distant countries within a configurable time window.
- After-hours login alerting - Rule targeting AAD logins outside defined business hours (07:00-19:00 local time).
- Unmanaged device rule - Alert when IsCompliantAndManaged is False combined with a non-UK country login.
- Anonymous link creation escalation - Higher severity rule for OneDrive AnonymousLinkCreated operations with LLM context on which files were shared.
- MISP threat intelligence integration - Replace hardcoded IP lists in rules with a CDB list automatically updated from a centralised MISP instance, enabling real-time threat intelligence across all client tenants without per-server API rate limiting.
- LLM tool-calling - Expand the Ollama integration to give the LLM access to tools (SIEM query, user history lookup) enabling it to autonomously gather additional context before drafting the ticket.
- Freshdesk custom fields - Map alert severity, rule ID, and affected workload to Freshdesk custom fields to enable SLA-based routing and reporting.

Conclusion

This project successfully delivered a functioning end-to-end SOC automation workflow from raw M365 log ingestion to analyst-ready Freshdesk tickets, validated in a homelab environment that mirrors the production configuration. The solution demonstrates that a significant portion of first-line alert triage can be automated without sacrificing analyst visibility or control.

The Knowledge Base-driven suppress/escalate logic provides a practical mechanism for reducing noise from known-good behaviour, while the LLM enrichment layer adds contextual analysis that would otherwise require analyst time to compile manually. The modular architecture of the active response script means additional enrichment sources, workloads, or ticketing integrations can be added incrementally.

The homelab environment is now a reliable platform for developing, testing, and validating detection rules and automation logic before deployment to production client tenants.